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Department	Computer Science and Engineering	**Program**	M.Tech.
Status		Active	
Program Category	Research Assistant - Research Assistantship		
Program validity date	2027-07-15	**Semester validity date?**	2027-01-31
Faculty Advisor(a)	Shivaram Kalyanakrishnan		

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Academic Performance Summary

Year	Sem	SPI	CPI	Sem Credits Used for SPI	Completed Semester Credits	Cumulative Credits Used for CPI	Completed Cumulative Credits
2025	Spring	10.0	9.15	16.0	16.0	54.0	54.0
2025	Autumn	10.0	8.79	6.0	6.0	38.0	38.0
2024	Spring	9.0	8.56	12.0	12.0	32.0	32.0
2024	Autumn	8.3	8.3	20.0	20.0	20.0	20.0

Semesterwise Details

*This registration is subject to approval(s) from faculty advisor/Course Instructor/Academic office.

Year/Semester: 2025-26/Spring

Course Code	Course Name	Tag	Credits	Grade	Crdit/Audit
CS 692	R & D Project II	Department elective	6.0	AA	C
CS 694	Seminar	Core course	4.0	AA	C
CS 728	Organization of Web Information	Department elective	6.0	AA	C
CS 899	Communication Skills	Core course	6.0	PP	N

Year/Semester: 2025-26/Autumn

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 601	Algorithms and Complexity	Additional Learning	6.0	BB	C
CS 691	R & D Project	Department elective	6.0	AA	C
CS 772	Deep Learning for Natural Language Processing	Additional Learning	6.0	AB	C

Year/Semester: 2024-25/Spring

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 695	Topics in Virtualization and Cloud Computing	Department elective	6.0	AB	C
CS 736	Medical Image Computing	Department elective	6.0	AB	C

Year/Semester: 2024-25/Autumn

Course Code	Course Name	Tag	Credits	Grade	Cr
CS 699	Software Lab.	Core course	8.0	BB	C
CS 725	Foundations of Machine Learning	Department elective	6.0	AB	C
CS 744	Design and Engineering of Computing Systems	Department elective	6.0	BB	C
GC 101	Gender in the workplace	Core course	0.0	PP	N
TA 101	Teaching Assistant Skill Enhancement & Training (TASET)	Core course	0.0	PP	N

Personal Info

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Emergency Contact	HEMANT SHINGEWAR, 9082770400 B-205, SUNRISE GALAXY CHS SANTOSHI MATA ROAD, RAMBAUG-4, KALYAN(W) 421301
Local Guardian	HEMANT SHINGEWAR, 9082770400 B-205, SUNRISE GALAXY CHS SANTOSHI MATA ROAD, RAMBAUG - 4, NEAR VIJAY SALES MAHARASHTRA Kalyan 421301

Student generated forms

1. Student Declaration Form
2. Hostel Registration Form
3. Medical File Form + Immunisation record

CS 728 Programming Assignment 3
Retrieval, Attention, and LLM

Utkarsh Verma
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1 Introduction

Retrieval is a core component in many modern NLP systems. While classical methods such as BM25 rely on lexical matching, and embedding-based methods capture semantic similarity, recent work suggests that attention mechanisms in large language models may also encode retrieval signals.

In this assignment, we explore retrieval using attention. We first evaluate traditional methods (Part 1), then directly use attention for retrieval (Part 2), and finally improve performance by selecting a subset of important attention heads (Part 3).

2 Part 1: Baseline Retrieval Methods

We evaluate three standard retrieval methods.

Method	Recall@1	Recall@5
BM25	0.1860	0.3476
MiniLM	0.3882	0.6106
UAE	**0.6044**	**0.8328**

Table 1: Baseline retrieval performance

Dense embedding methods clearly outperform lexical methods, with UAE achieving the best performance.

3 Part 2: Attention-based Retrieval

In Part 2, we use attention scores to rank tools.

Metric	Value
Recall@1	0.0122
Recall@5	0.2748

Table 2: Attention-based retrieval using all heads

The performance is significantly lower than embedding methods, indicating that using all attention heads introduces substantial noise.

Attention vs Position

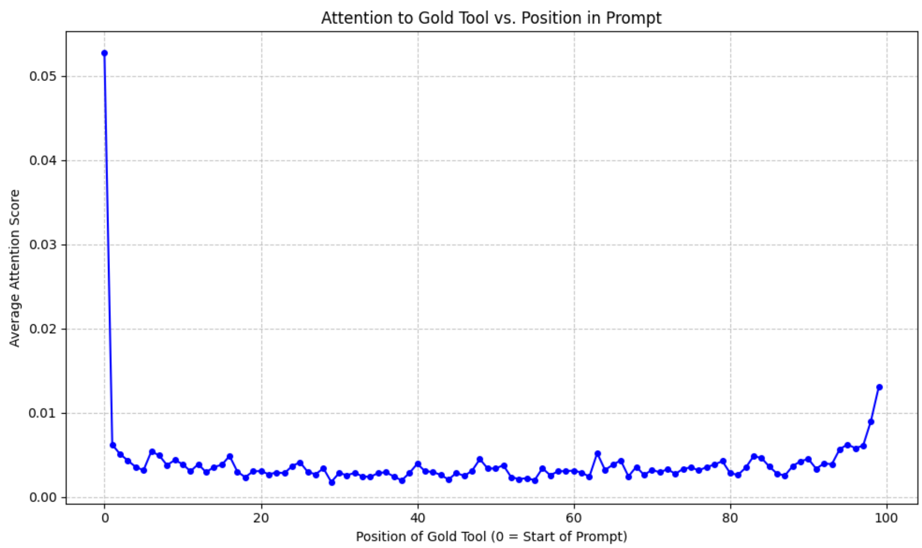


Figure 1: Attention vs position using all heads

4 Part 3: Retrieval using Selected Heads

4.1 Head Selection Strategies

In order to identify the most useful attention heads for retrieval, we explore different ways of assigning scores to each head based on its performance on training queries. The underlying idea is that a good head should consistently assign higher relevance to the correct tool.

We experiment with three strategies.

Reciprocal Rank (MRR-based) In this approach, each head is scored using the reciprocal of the rank of the correct tool:

$$\text{score} = \frac{1}{\text{rank} + 1}$$

This formulation is inspired by the Mean Reciprocal Rank (MRR) metric, which is widely used in information retrieval. MRR evaluates how close the correct answer appears in a ranked list, assigning higher weight to top positions. A head that consistently ranks the correct tool near the top will therefore accumulate a higher score.

A key advantage of this method is that it provides a *graded reward*. For example, moving the correct tool from rank 5 to rank 2 significantly increases the score. This allows the method to capture not only perfect predictions but also near-correct ones.

Recall@1-based Scoring In this strategy, a head is rewarded only if it ranks the correct tool at the top:

$$\text{score} = \begin{cases} 1 & \text{if rank} = 0 \\ 0 & \text{otherwise} \end{cases}$$

This is a stricter criterion compared to reciprocal rank. While it directly aligns with the Recall@1 metric, it ignores useful information from near-correct rankings, which can make the selection process less stable.

Gold Attention Score This method scores each head based on the raw attention it assigns to the correct tool:

$$\text{score} = \text{attention}(\text{query} \rightarrow \text{gold tool})$$

Unlike the previous methods, this approach does not explicitly consider ranking. While it measures how much attention a head gives to the correct tool, it may not reflect whether the tool is actually ranked higher than others.

Final Decision Among these strategies, Reciprocal Rank provides a good balance between strictness and flexibility. It aligns naturally with ranking-based evaluation metrics and encourages heads that consistently improve the position of the correct tool, rather than focusing only on exact matches or raw attention values.

4.2 Effect of Number of Heads (MRR)

Here, we present the comparison of different numbers of attention heads selected for the chosen approach-

Heads	Recall@1	Recall@5
10	**0.4044**	**0.7036**
20	0.3558	0.6702
30	0.2796	0.6272

Table 3: Effect of number of heads using MRR

4.3 Comparison of All Strategies

Here, we present a tabular comparison of the results of all the strategies used in part 3-

Strategy	Heads	Recall@1	Recall@5
MRR	10	**0.4044**	**0.7036**
MRR	20	0.3558	0.6702
MRR	30	0.2796	0.6272
Recall@1	10	0.3990	0.7036
Recall@1	20	0.3526	0.6684
Gold Score	10	0.1236	0.6728
Gold Score	20	0.1414	0.6084

Table 4: Comparison of all strategies across different number of heads

4.4 Comparison with Part 2

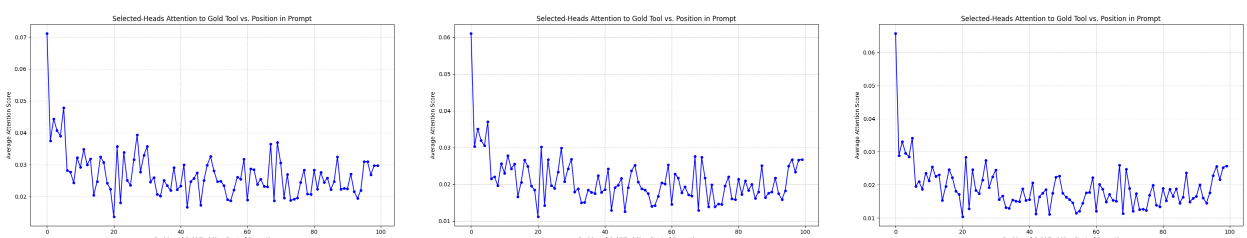
Here, we present a tabular comparison of part 2 results and part 3 results-

Method	Recall@1	Recall@5
Part 2 (All Heads)	0.0122	0.2748
Part 3 (Best: MRR, 10 heads)	**0.4044**	**0.7036**

Table 5: Improvement from head selection

4.5 Visualization

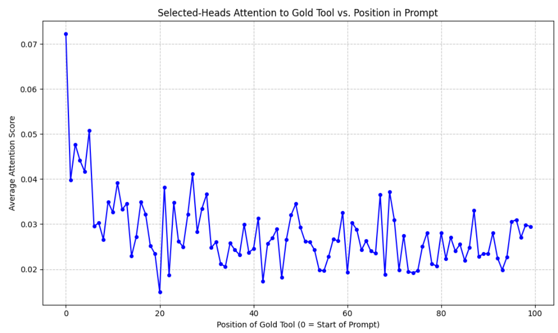
MRR (Different Heads)



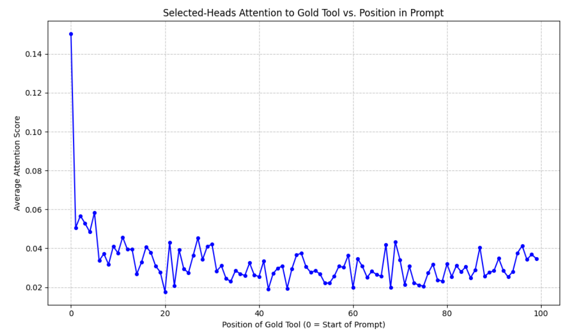
(a) 10 heads (b) 20 heads (c) 30 heads

Figure 2: MRR-based attention vs position for different head counts

Other Strategies



(a) Recall@1



(b) Gold Score

Figure 3: Comparison of different head selection strategies

5 Analysis

5.1 Why Head Selection Works

Using all heads leads to poor performance because many heads capture irrelevant patterns. By selecting only the most useful heads, we remove noise and focus on meaningful signals.

5.2 Effect of Number of Heads

Performance is highest with 10 heads and decreases as more heads are added. This shows that only a small subset of heads is important, and adding more introduces noise.

5.3 Comparison of Strategies

MRR consistently performs best, followed closely by Recall@1. Gold Score performs significantly worse.

5.4 Why Reciprocal Rank Works

MRR uses:

$$\frac{1}{\text{rank} + 1}$$

This gives higher reward when the correct tool is ranked near the top.
A simple justification comes from Mean Reciprocal Rank:

$$\text{MRR} = \frac{1}{Q} \sum \frac{1}{\text{rank}_i}$$

For example:

- $\text{rank} = 1 : \text{score} = 1$
- $\text{rank} = 5 : \text{score} = 0.2$

Thus, improving rank directly increases the score. Unlike Recall@1, this method does not ignore near-correct predictions, making it more stable.

5.5 Attention vs Position

Attention generally decreases as the position increases. However, selected heads are less sensitive to position, indicating that important heads capture more meaningful relationships.

6 Conclusion

We show that attention alone is not sufficient for retrieval when all heads are used. However, selecting a small subset of heads dramatically improves performance.

Among the strategies, Reciprocal Rank works best because it balances strictness and flexibility. These results suggest that attention heads are specialized, and only a few of them are responsible for retrieval.